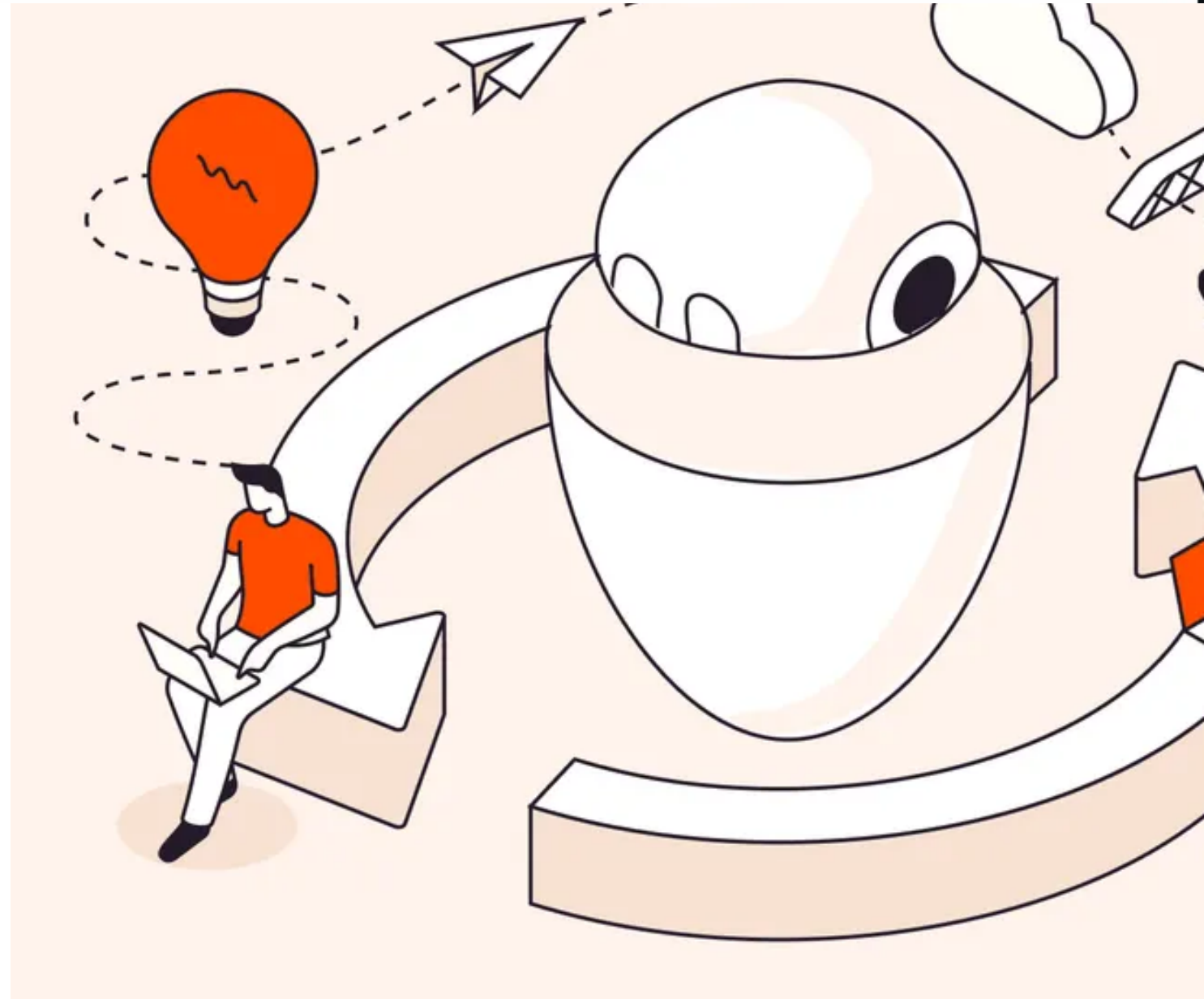


Week 1 – Lecture 2

# AI Lifecycle & Real-World Applications

Artificial Intelligence Foundations

 CSE276 | Unit I



# Quick Recap of Lecture 1



## What is AI?

Machines mimicking human intelligence.



## AI History

From Turing Test to Generative AI.



## Types of AI

Narrow AI (ANI) vs General (AGI).



## AI vs ML vs DL

Nested subsets of technology.

# A Simple Question



Netflix recommends  
a movie



Google Maps finds  
a faster route

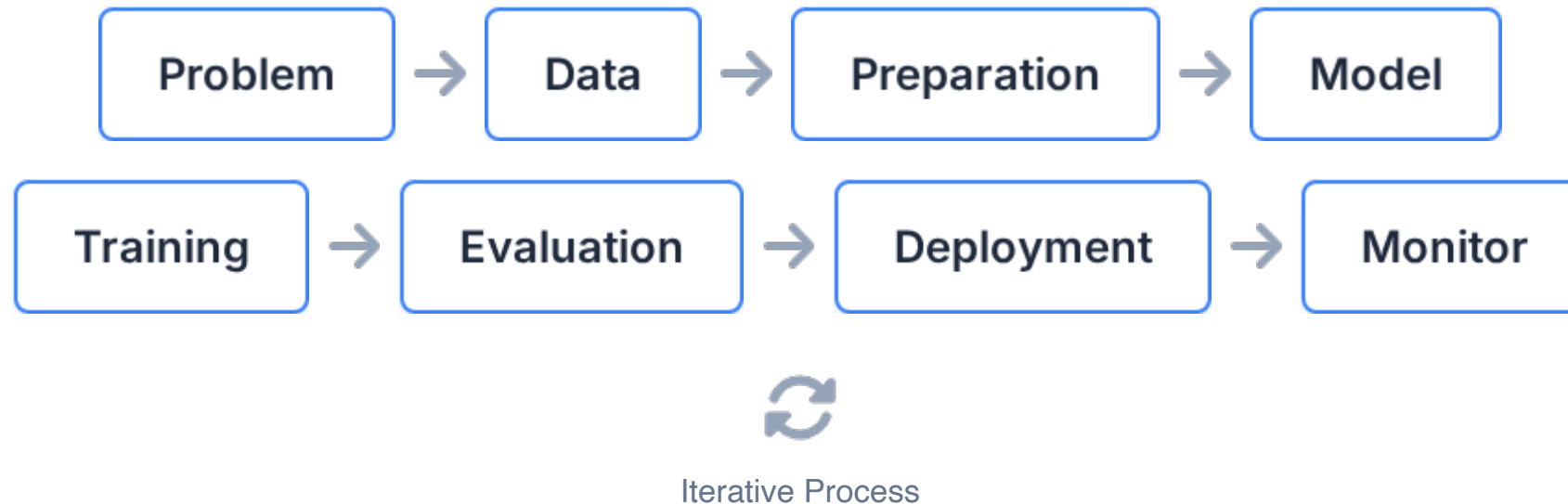


Phone unlocks  
using your face

**"How do these AI systems actually get built?"**

# What is an AI Lifecycle?

Building AI is a structured process, not just writing code.





# Step 1 – Problem Definition

Every AI project begins with solving a real problem.



Spam detection



Disease diagnosis



Predicting rainfall



Credit card fraud

## Analogy

*"A doctor doesn't prescribe medicine before understanding the illness."*

## | Step 2 – Data Collection



Images



Audio



Videos



Sensors



Documents



Databases

**"AI is only as good as its data."**

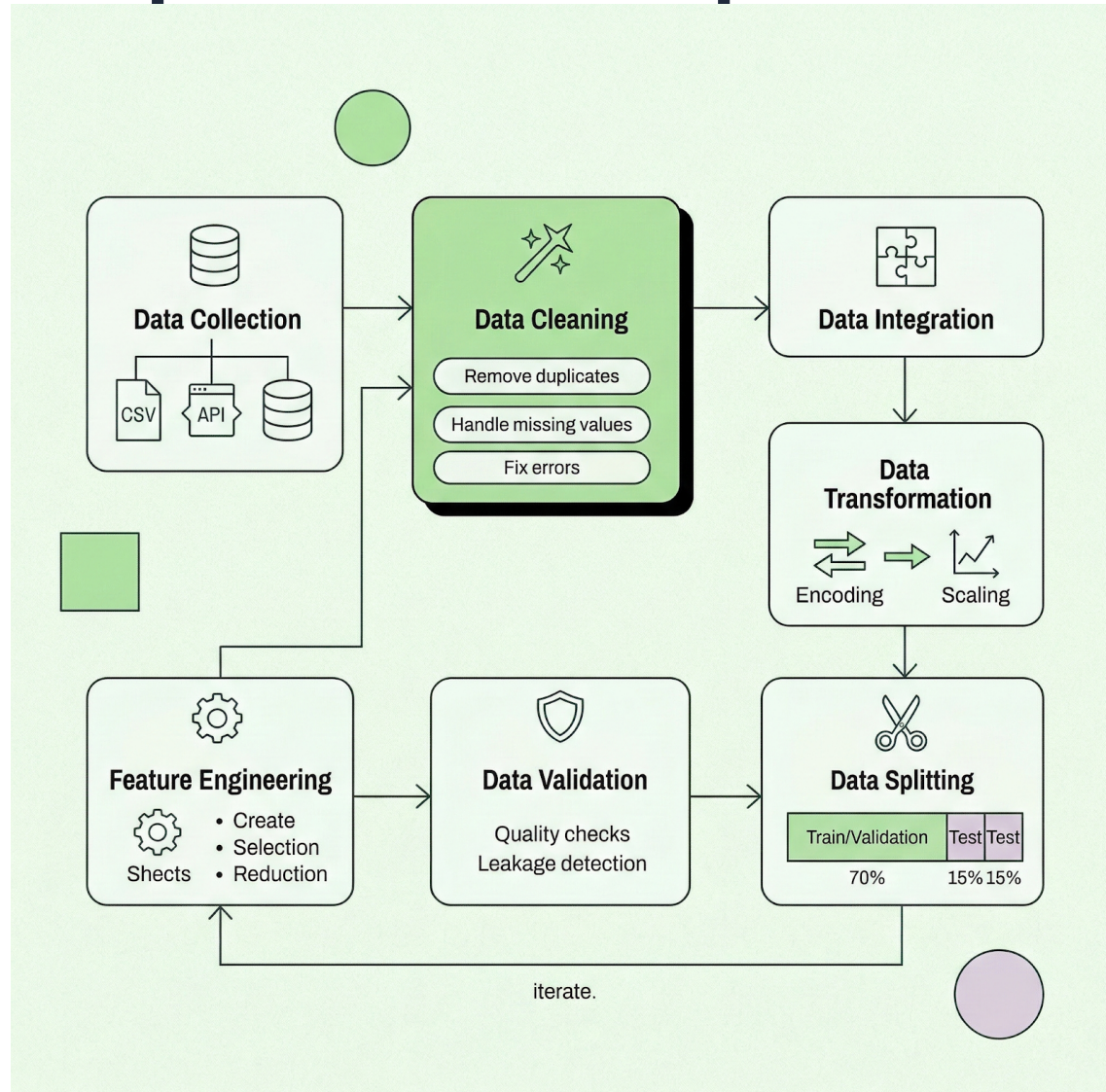


# Interactive Activity

"If you wanted to build an AI that predicts student grades, what data would you collect?"

DISCUSS WITH YOUR  
NEIGHBOR

# Step 3 – Data Preparation



⚠ Handling missing values

📄 Removing duplicates

🏷 Fixing incorrect labels

📄 Standardizing formats

## 🍴 Analogy

"Preparing ingredients before cooking."

# Step 4 – Model Selection

Different problems require different AI models.



## Classification

Categorizing data (e.g.,  
Spam or Not Spam)



## Regression

Predicting numbers (e.g.,  
House Prices)



## Clustering

Grouping similar items  
(e.g., Customer  
Segments)



## Neural Networks

Complex patterns (e.g.,  
Image Recognition)

## Step 5 – Training the Model



### Analogy

"Teaching a student using many practice questions."





## Step 6 – Testing & Evaluation

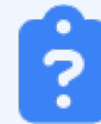
Accuracy on seen data is not enough.

### Training Data



Data the model has already "studied".

### New / Unseen Data



The real test of generalization.

# Interactive Poll

"If an AI scores 100% during training but performs poorly in real life, is it a good AI system?"

**A) Yes, 100% is perfect.**

**B) No, it memorized instead of learning.**

**C) It depends on the programming language.**



# Interactive Poll

"If an AI scores 100% during training but performs poorly in real life, is it a good AI system?"

A) Yes, 100% is perfect.



B) No, it memorized instead of learning.

C) It depends on the programming language.

# Step 7 – Deployment



Users only interact with the deployed model.



Mobile Apps



Websites



Hospitals



Cars



Banks

## Step 8 – Monitoring & Improvement

 AI models degrade over time.

 Data trends change constantly.

 User behavior evolves.

### Example: Spam Filters

Spammers invent new tactics; the AI must be re-trained on new spam emails to stay effective.



### The Continuous Loop



# The Complete AI Lifecycle & Roles



| Lifecycle Stages        | Key Professionals Involved           |
|-------------------------|--------------------------------------|
| Problem Definition      | Domain Experts, Product Managers     |
| Data Collection & Prep  | Data Engineers, Data Analysts        |
| Model Selection & Train | Data Scientists, ML Engineers        |
| Deployment & Monitor    | Software Developers, MLOps Engineers |

# Real-World Applications of AI



Healthcare



Finance



Education



Agriculture



Transportation



Manufacturing



Entertainment



Cybersecurity



# AI Across Industries

| Domain        | AI Application         | Benefit          | Example                         |
|---------------|------------------------|------------------|---------------------------------|
| Healthcare    | Medical Diagnosis      | Early detection  | Detecting tumors in X-rays      |
| Finance       | Fraud Detection        | Security         | Blocking suspicious card swipes |
| Education     | Personalized Learning  | Better outcomes  | Adaptive tutoring systems       |
| Agriculture   | Crop Disease Detection | Higher yield     | Scanning leaves via drone       |
| Manufacturing | Predictive Maintenance | Reduced downtime | Predicting machine failure      |
| Entertainment | Movie Recommendations  | Engagement       | Netflix homepage                |

# Case Study: How Google Maps Uses AI

1. GPS Data Collection



2. Real-time Traffic Info



3. AI ETA Prediction



4. Route Optimization



5. User Navigation





# Board Work

Draw the AI lifecycle.

"Let's place our real-world examples (Spam, Netflix, Maps) into the correct stages."





# | Key Takeaways



AI starts with  
a problem.



Data drives  
AI.



Models learn  
from data.



Continuous  
improvement.



AI powers  
industries.

# Quick Quiz

**What is the primary reason an AI model is evaluated on "Unseen Data"?**

- ☐ A) To collect more data.
- ☐ B) To ensure it generalized, not just memorized.
- ☐ C) To increase training speed.
- ☐ D) It is not evaluated on unseen data.

# Quick Quiz

**What is the primary reason an AI model is evaluated on "Unseen Data"?**

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# Preview: Unit II Formulation

## Formulating an AI Problem

-  State-space
-  Search space
-  Solution space
-  Problem characteristics



# Image Sources

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